Vijayaraj UA

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Summary — MERN Full Stack Developer with 3+ years of production experience shipping 5 enterprise platforms across healthcare, e-learning, live commerce, and collaboration verticals. Engineered WebRTC multi-user video conferencing (SFU/Mediasoup) and Socket.IO real-time systems from scratch. Leverages AI-assisted development workflows to compress delivery cycles by 40–50%. Strong command of both SQL and NoSQL architectures; writes clean, scalable, event-driven code that performs under concurrent load.

Skills

- **Languages**: JavaScript (ES6+), TypeScript (basics)
- **Frontend**: React.js, Redux, Context API, React Hooks, HTML5, CSS3, Responsive Design, Component Architecture
- **Backend**: Node.js, Express.js, GraphQL, REST APIs, JWT Authentication, RBAC, Microservices, MongoDB Aggregation Pipeline, Data Modelling
- **Real-Time**: WebRTC, Socket.IO, WebSockets, Presence Tracking
- **Databases**: MongoDB, MySQL / SQL, DynamoDB, Redis, Mongoose ODM
- **Cloud & DevOps**: AWS (EC2, S3, Cognito, CloudWatch, Serverless), Git, CI/CD basics
- **Architecture**: Microservices, Event-Driven Systems, API Design, Role-Based Access Control
- **Tools & Productivity**: Git, Postman
- **AI Tools**: Cursor, Windsurf, Antigravity, Claude Code, Github Co-pilot

Experience

Insta VC **JUN 2024 – JUN 2026**
Software Developer (MERN Stack)

- Delivered 5 production-grade full-stack platforms in under 18 months across healthcare, e-learning, live commerce, and enterprise collaboration verticals – each shipped to real paying customers.
- Cut feature development time by 40–50% by integrating AI coding tools into daily workflow, enabling single-handedly owning modules that would typically require 2–3 engineers.
- Architected a WebRTC video conferencing engine using Mediasoup SFU, supporting concurrent multi-user audio/video streams with role-separated workflows (Admin / Agent / Customer).
- Built a WhatsApp-style end-to-end encrypted chat system with Socket.IO delivering sub-100ms message latency, online/offline presence, typing indicators, and read receipts at scale.
- Designed a RESTful + GraphQL API layer handling JWT authentication, RBAC, and license-gated feature access – reducing unauthorized access incidents to zero post-launch.
- Optimised event-driven Node.js backend to sustain high-volume concurrent WebSocket connections, improving throughput and cutting response times for real-time operations.

NINEX CORP **JUL 2026 – Present**
MERN Stack Developer

- Engineered features for an **MT5-style real-time trading platform** using **Next.js, Bun, Elysia, WebSockets, Redis, and Kafka**, supporting real-time market data, trading operations, account management, and multi-role workflows. “‘
- Designed and implemented **event-driven and real-time backend services**, leveraging Kafka for asynchronous event processing, Redis for caching and fast data access, and WebSockets for low-latency client updates.
- Developed scalable **RESTful APIs and trading services** with Bun and Elysia, implementing JWT authentication, RBAC, vendor/user management, trading accounts, orders, settlements, and platform administration workflows.
- Optimized real-time communication and backend processing to support **high-concurrency WebSocket connections** and responsive trading workflows across multiple connected clients.
- Built production-grade full-stack applications across healthcare, e-learning, live commerce, and enterprise collaboration, owning features across **React/Next.js, Node.js, MongoDB, WebSockets, and WebRTC**.
- Architected a **WebRTC conferencing platform using Mediasoup SFU** and developed a WhatsApp-style real-time chat system with Socket.IO, including presence, typing indicators, read receipts, and low-latency messaging. “‘

Key Projects

InMeet – Enterprise Meeting & Chat Platform **2025**
Slack + Zoom alternative for enterprise teams; real-time collaboration at scale

- Engineered the full Meetings module from scratch – video rooms, scheduling, participant management, and recording hooks – using WebRTC + Mediasoup SFU.
- Built a production WhatsApp-style chat system with E2E encryption and Socket.IO presence (online/offline + typing indicators), sustaining thousands of concurrent connections.
- Designed horizontally scalable MongoDB schemas and Node.js services to handle high-volume concurrent users without performance degradation.

InClinic – Healthcare Management Platform **2025**
End-to-end clinic OS: multi-role patient/doctor/admin workflows with live scheduling

- Eliminated appointment booking conflicts by engineering a real-time slot-locking system using WebSocket events and atomic MongoDB writes.
- Shipped multi-currency payment integration with automatic location-based currency detection, enabling clinics to serve patients across regions without manual configuration.
- Implemented an RBAC model (Admin / Doctor / Patient) with JWT authentication, prescription management, and audit-trail logging aligned to healthcare data practices.

InClass – Online Learning Management System **2024**
School management SaaS: classes, assignments, tests, results, real-time notifications

- Delivered a full LMS (assignments, tests, grading, class management) with role-differentiated dashboards for teachers and students built on React.js + Node.js + MongoDB.
- Reduced notification latency to near-zero by replacing a polling mechanism with socket-based push for new assignments and results.
- Built a responsive, mobile-first UI in React.js, eliminating layout breakage across device sizes reported in QA.

InShop – Live Video Shopping Platform **2024**
Real-time jewelry showcase: live video + chat + product interaction

- Built a live commerce platform enabling jewellery agents to conduct WebRTC video sales sessions with customers, integrating simultaneous real-time product browsing and chat.
- Delivered full role-based workflow (Admin / Agent / Customer) with feature-gated access and live session management – end-to-end in under 3 months.

Digital Signage System **2024**
Multi-screen dynamic content display with license-based access control

- Built a content distribution platform enabling real-time screen updates across multiple endpoints simultaneously, with RBAC and license-tier configurations for enterprise clients.

Education

SRM Valliammai Engineering College
B. Tech – Artificial Intelligence and Data Science — CGPA: 8.5 *2020–2024*
Chengalpattu, Tamil Nadu